A Stochastic Structural Framework for Measuring Implied Bitcoin Accretion in MicroStrategy Equity: Corrected Model and Implementation Plan

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1 Objectives and Scope

The goal of this project is to:

- Build a mathematically consistent structural model linking Bitcoin (BTC) price dynamics, MicroStrategy (MSTR) BTC holdings, its balance-sheet leverage, and the equity premium over net asset value (NAV).
- Distinguish between:
  - *Structural balance-sheet leverage* (from assets and debt).
  - *Total equity elasticity* (including the response of the premium to BTC).
  - *Total BTC growth* versus *per-share BTC accretion*.
- Calibrate the model to publicly available data.
- Compute and visualize:
  - Implied BTC Growth Rate (IBGR) – total and per-share.
  - Implied Leverage Elasticity (ILE).
  - Total Equity Elasticity (TEE).
  - Premium Mean-Reversion Index (PMRI).
  - Implied Funding Requirement Distribution (IFRD).
  - Survival / recapitalization probabilities.
- Implement the full pipeline in Python with clear data ingestion, calibration, simulation, and reporting.

2 Model Specification

2.1 State Variables, Measures, and Balance Sheet

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0})$ with two measures:

- $\mathbb{P}$: historical (real-world) measure, used for premium dynamics, management behavior (holdings), and survival/IFRD analysis.

- $\mathbb{Q}$: risk-neutral measure, used (if needed) to calibrate BTC volatility/jump parameters to options.

State variables:

- S_t: BTC (BTC-USD) price.
- H_t: MicroStrategy BTC holdings (in BTC).
- D_t: outstanding debt (face-value), modeled deterministically at baseline.
- A_t: BTC asset value,
$$A_t = H_t S_t.$$
- P_t: MSTR share price.
- N_t^{sh}: number of MSTR shares outstanding.
- E_t: equity market value,
$$E_t = P_t N_t^{\mathrm{sh}}.$$
- B_t: BTC per share,
$$B_t = \frac{H_t}{N_t^{\mathrm{sh}}}.$$

Define net asset value (NAV) with limited liability:

$$\mathrm{NAV}_t = (A_t - D_t)^+ = \max(H_t S_t - D_t, 0).$$

When computing the historical log-premium from data we will:

- Introduce a small floor $\varepsilon > 0$ to avoid log singularities:
$$\mathrm{NAV}_t^{\mathrm{clip}} = \max(A_t - D_t, \varepsilon),$$
- and define
$$\pi_t = \log\left(\frac{E_t}{\mathrm{NAV}_t^{\mathrm{clip}}}\right).$$

Periods where $A_t \approx D_t$ will be tagged as a "distress regime" and treated carefully in calibration.

2.2 BTC Price Dynamics

Under the risk-neutral measure $\mathbb{Q}$, for option calibration we consider a jump-diffusion:

$$\frac{dS_t}{S_{t^-}} = (r_t - q_S - \lambda_S k_S)\,dt + \sigma_S(S_t, t)\,dW_t^{S,\mathbb{Q}} + \left(e^{Y_S} - 1\right)dN_t^S,$$

where:

- r_t is the risk-free rate (deterministic or piecewise constant).
- q_S is any BTC "carry" term (set $q_S = 0$ initially).
- $W_t^{S,\mathbb{Q}}$ is a Brownian motion under $\mathbb{Q}$.

- N_t^S is a Poisson process with intensity λ_S.
- Y_S is the (log-)jump size, with distribution ν_S.
- $k_S = \mathbb{E}[e^{Y_S} - 1]$ is the jump compensator.
- $\sigma_S(S_t, t)$ is a (possibly) state-dependent volatility surface calibrated to options.

For historical $\mathbb{P}$-measure simulations focusing on survival and IFRD, a simpler GBM with constant volatility fitted to BTC returns is acceptable as a first step:

$$\frac{dS_t}{S_t} = \mu_S dt + \sigma_S dW_t^{S,\mathbb{P}}.$$

2.3 Premium Dynamics under $\mathbb{P}$

We model the log-premium π_t as an Ornstein–Uhlenbeck (OU) process under $\mathbb{P}$ (management and sentiment world), with optional jumps:

$$d\pi_t = \kappa(\theta - \pi_t)dt + \sigma_\pi dW_t^{\pi,\mathbb{P}} + \eta_\pi dJ_t^\pi,$$

where:

- $\kappa > 0$ is the mean-reversion speed.
- θ is the long-run mean premium.
- σ_π is the volatility of premium.
- $W_t^{\pi,\mathbb{P}}$ is a Brownian motion under $\mathbb{P}$.
- J_t^π is a pure-jump process with intensity λ_π (can be set to zero initially).
- The Brownian motions are correlated:

$$\mathrm{corr}(dW_t^{S,\mathbb{P}}, dW_t^{\pi,\mathbb{P}}) = \rho.$$

Discrete-time approximation for daily step Δt (ignoring jumps initially):

$$\pi_{t+\Delta t} = \pi_t + \kappa(\theta - \pi_t)\Delta t + \sigma_\pi \sqrt{\Delta t}\, Z_\pi,$$

with (Z_S, Z_π) bivariate normal and $\mathrm{corr}(Z_S, Z_\pi) = \rho$.

2.4 BTC Holding Dynamics (Flywheel) under $\mathbb{P}$

We encode the "leverage flywheel" via an endogenous BTC holding process:

$$dH_t = \alpha\, \pi_t^+ dt + dH_t^{\text{jump}}, \quad \pi_t^+ = \max(\pi_t, 0),$$

where the jump component captures discrete financing events:

$$dH_t^{\text{jump}} = \sum_k \Delta H_{\tau_k}\, \delta_{t=\tau_k}, \qquad \Delta H_{\tau_k} \geq 0.$$

Here:

- $\{\tau_k\}$ are event times of a Poisson process M_t with intensity λ_M.
- ΔH_{τ_k} are BTC amounts purchased using financing proceeds.

In discrete time:

- Draw $N_t^M \sim \text{Poisson}(\lambda_M \Delta t)$.
- Draw jump sizes $\Delta H_{t,1}, \ldots, \Delta H_{t,N_t^M}$ from a distribution $\mathcal{D}_H$.
- Update

$$H_{t+\Delta t} = H_t + \alpha \pi_t^+ \Delta t + \sum_{i=1}^{N_t^M} \Delta H_{t,i}, \qquad H_{t+\Delta t} \geq 0.$$

Calibration choices:

- α is estimated on aggregated (weekly/monthly) holding changes to avoid daily noise.
- If data suggests $\alpha \approx 0$, we can set $\alpha = 0$ and model all accretion as jumps, potentially with λ_M depending on the premium (e.g. $\lambda_M(\pi_t)$).
- As an extension, the jump intensity or sizes can be made to depend on leverage (e.g. $\beta_{LS}(t)$) to capture over-leverage constraints.

2.5 Debt Process and Convertibles

At baseline, we treat total debt as deterministic piecewise-constant:

$$D_t = \sum_{j=1}^{J} D_j \mathbf{1}_{\{t<T_j\}},$$

where each instrument j has face value D_j and maturity T_j. This is conservative for solvency analysis, especially for convertible notes (we ignore their equity-like behavior when deep in the money).

In implementation, each instrument will be represented by a `DebtInstrument` object with fields:

- Face value, coupon, issue date, maturity date.
- Convertible flag, conversion price, and other relevant terms.

This allows an extension where an "effective debt" D_t^{eff} discounts the equity-like component of deep in-the-money convertibles.

2.6 Share-Count Dynamics and BTC per Share

To distinguish total BTC growth from per-share accretion, we model the share count:

$$dN_t^{\text{sh}} = dN_t^{(\text{equity})} + dN_t^{(\text{conv})},$$

where:

- $dN_t^{(\text{equity})}$ captures new stock issuance (e.g. ATMs, follow-ons).

- $dN_t^{(\text{conv})}$ captures conversion of convertible notes.

In practice, both components are lumpy and are best modeled as jump processes with event times aligned with financing and conversion events.

The BTC per share process is:

$$B_t = \frac{H_t}{N_t^{\text{sh}}}.$$

Per-share accretion metrics will be defined in terms of B_t.

2.7 Equity Valuation, Structural Leverage, and Total Elasticity

The structural relation between equity, assets, debt, and premium is:

$$E_t = e^{\pi_t} \cdot (A_t - D_t)^+ = e^{\pi_t}(H_t S_t - D_t)^+.$$

Implied Leverage Elasticity (ILE). Define the *structural* leverage elasticity as the sensitivity of equity to BTC price, ignoring instantaneous changes in premium:

$$\beta_{LS}(t) = \left.\frac{\partial \log E_t}{\partial \log S_t}\right|_{\pi_t \text{ fixed}} \approx \frac{H_t S_t}{H_t S_t - D_t},$$

whenever $H_t S_t > D_t$. This depends only on the current balance sheet and is independent of π_t.

Total Equity Elasticity (TEE). In reality, π_t typically depends on BTC shocks. From

$$\log E_t = \pi_t + \log(H_t S_t - D_t),$$

one obtains

$$\frac{d \log E_t}{d \log S_t} = \frac{d\pi_t}{d \log S_t} + \frac{H_t S_t}{H_t S_t - D_t}.$$

Define the *Total Equity Elasticity*

$$\beta_{\text{TOT}}(t) = \frac{d \log E_t}{d \log S_t} \approx \gamma_{\pi S} + \beta_{LS}(t),$$

where $\gamma_{\pi S}$ is the empirical sensitivity

$$\gamma_{\pi S} \approx \frac{\text{Cov}(\Delta \pi_t, \Delta \log S_t)}{\text{Var}(\Delta \log S_t)},$$

estimated via regression. We will report both $\beta_{LS}(t)$ (pure balance-sheet leverage) and $\beta_{\text{TOT}}(t)$ (what traders *feel* in equity P&L).

3 Implied Quantitative Indicators

3.1 Implied BTC Growth Rate (IBGR)

Total IBGR. Under the historical measure $\mathbb{P}$, define the expected total BTC holding growth over a short horizon Δt:

$$\mu_H = \frac{1}{H_0} \mathbb{E}^{\mathbb{P}} \left[\frac{H_{\Delta t} - H_0}{\Delta t} \,\middle|\, S_0, \pi_0, H_0 \right].$$

Using the holding dynamics for small Δt,

$$\mu_H \approx \frac{1}{H_0} \left(\alpha \pi_0^+ + \lambda_M \mathbb{E}[\Delta H] \right),$$

where $\mathbb{E}[\Delta H]$ is the mean BTC added per financing event.

Per-Share IBGR. Define BTC per share $B_t = H_t/N_t^{\text{sh}}$. Per-share accretion is captured by

$$\mu_{H/N} = \frac{1}{B_0}\mathbb{E}^{\mathbb{P}}\left[\frac{B_{\Delta t} - B_0}{\Delta t}\,\middle|\, S_0, \pi_0, H_0, N_0^{\text{sh}}\right].$$

Exact closed form is messy; in practice we will:

- Simulate joint paths of (H_t, N_t^{sh}) under the calibrated dynamics.
- Estimate $\mu_{H/N}$ via Monte Carlo.

Positive $\mu_{H/N}$ indicates *accretive* BTC growth per share.

3.2 Implied Leverage Elasticity (ILE)

As above, for a given state (S_t, H_t, D_t),

$$\text{ILE}_t = \beta_{LS}(t) = \frac{H_t S_t}{H_t S_t - D_t},$$

whenever $H_t S_t > D_t$. This is purely a function of balance-sheet leverage.

3.3 Total Equity Elasticity (TEE)

Given the empirical $\gamma_{\pi S}$ and current $\beta_{LS}(t)$:

$$\beta_{\text{TOT}}(t) \approx \gamma_{\pi S} + \beta_{LS}(t).$$

This indicator captures the full sensitivity of equity to BTC, including premium dynamics.

3.4 Premium Mean-Reversion Index (PMRI)

For the OU process (ignoring jumps), the stationary variance is

$$\text{Var}^{\text{ss}}(\pi_t) = \frac{\sigma_\pi^2}{2\kappa}.$$

Define the z-score:

$$\text{PMRI} = \frac{\pi_0 - \theta}{\sigma_\pi/\sqrt{2\kappa}},$$

with large positive values indicating exuberant premium, and negative values indicating discount/stress.

3.5 Implied Funding Requirement Distribution (IFRD)

The cumulative dollar cost of BTC acquisitions over horizon Δt is

$$G_{\Delta t} = \int_0^{\Delta t} S_u \, dH_u.$$

Using the decomposition:

$$G_{\Delta t} = \int_0^{\Delta t} S_u \alpha \pi_u^+ \, du + \sum_{0<u\le\Delta t} S_u \Delta H_u.$$

In discrete form for simulated paths:

$$G_{\Delta t}^{(\text{path})} \approx \sum_{k=0}^{n-1} S_{t_k}(H_{t_{k+1}} - H_{t_k}).$$

Monte Carlo over many paths provides the empirical distribution of $G_{\Delta t}$: mean, variance, and VaR/ES (e.g. $\text{VaR}_{95\%}(G_{\Delta t})$).

3.6 Survival and Recapitalization Probabilities

Define a distress threshold with buffer $\varepsilon \geq 0$:

$$A_t \leq (1+\varepsilon)D_t \quad \Longleftrightarrow \quad H_t S_t \leq (1+\varepsilon)D_t.$$

Let

$$\tau = \inf\{u \in [0,T] : A_u \leq (1+\varepsilon)D_u\}$$

be the first-passage time to distress.

Simulating under $\mathbb{P}$:

- The survival probability is

$$\mathbb{P}(\tau > T) \approx \frac{\#\{\text{paths with no hit before } T\}}{\#\{\text{total paths}\}}.$$

- Recapitalization probabilities can be defined with thresholds on equity, premium, or funding requirements.

4 Data Requirements and Download Instructions

All series should be aligned to a common daily date index.

4.1 BTC Price and Derivatives

BTC Spot/OHLCV.

- BTC-USD daily close (or OHLCV) series S_t.
- Sources:
  - Yahoo Finance (`"BTC-USD"`), via web or `yfinance`.
  - CryptoDataDownload / CoinGecko / CoinMarketCap for CSV downloads.

BTC Options (Optional). For a richer volatility/jump calibration under $\mathbb{Q}$:

- Deribit BTC options chain.
- CME BTC futures/options from a data vendor.

4.2 MSTR Equity Data

- Daily MSTR share price P_t.
- Shares outstanding N_t^{sh} (quarterly + event-based).
- Sources:
  - Yahoo Finance (`"MSTR"`) for price.
  - SEC filings (10-K, 10-Q), MSTR investor relations, data sites for share count history.

4.3 MicroStrategy BTC Holdings

- BTC holdings time series H_t (in BTC).
- Purchase events table: date, BTC added, USD notional, average price.
- Sources:
  - MicroStrategy / company site disclosures.
  - Corporate BTC tracking sites with CSV downloads.

4.4 Debt Schedule and Financing Events

- Debt instruments: issue date, maturity, face value, coupon, convertible terms.
- Financing events: equity issuance, new debt, conversions.
- Sources:
  - SEC EDGAR (10-K, 10-Q, 8-K).
  - Company press releases / investor presentations.

4.5 Risk-Free Curve

- US Treasury yields (e.g. 1Y, 2Y) from FRED.
- Simple version: use a constant annual risk-free rate (e.g. 3%).

5 Estimation and Calibration Plan

5.1 Data Ingestion and Alignment

1. Download BTC, MSTR price series, holdings, debt schedule, shares outstanding as CSV or via APIs.
2. In Python (`pandas`), build a daily date index from first BTC purchase to latest data.
3. Merge:
   - Forward-fill H_t, D_t, and N_t^{sh} between known dates.
   - For non-trading days for MSTR, forward-fill P_t or work on trading-day-only calendar.

4. Compute derived series:
   - $A_t = H_t S_t$.
   - $\mathrm{NAV}_t = (A_t - D_t)^+$.
   - $\mathrm{NAV}_t^{\mathrm{clip}} = \max(A_t - D_t, \varepsilon)$.
   - $E_t = P_t N_t^{\mathrm{sh}}$.
   - $\pi_t = \log(E_t/\mathrm{NAV}_t^{\mathrm{clip}})$.
   - $B_t = H_t/N_t^{\mathrm{sh}}$.
5. Tag dates with $A_t \approx D_t$ (or NAV_t close to zero) as distress regime indicators.

5.2 Calibration of Premium OU under $\mathbb{P}$

1. Use daily (or business-day) steps $\Delta t = 1/252$.
2. Exclude distress-regime datapoints or treat them separately.
3. Use the exact OU transition:

$$\pi_{t+\Delta t} = \theta + (\pi_t - \theta)e^{-\kappa \Delta t} + \sigma_\pi \sqrt{\frac{1 - e^{-2\kappa \Delta t}}{2\kappa}}\, Z_t,$$

   and perform maximum likelihood or nonlinear least squares to estimate $(\kappa, \theta, \sigma_\pi)$.
4. Estimate ρ by empirical correlation between BTC log returns $\Delta \log S_t$ and premium changes $\Delta \pi_t$.
5. Estimate $\gamma_{\pi S}$ via regression of $\Delta \pi_t$ on $\Delta \log S_t$ to obtain total elasticity.

5.3 Calibration of BTC Holding Dynamics

1. Construct ΔH_t (changes in holdings) and identify financing event dates from disclosures.
2. Build weekly or monthly aggregates:

$$\Delta H_t^{(W)}, \quad \overline{\pi_t^+}^{(W)}.$$

3. On non-event periods, regress:

$$\Delta H_t^{(W)} \approx \alpha \overline{\pi_t^+}^{(W)} \Delta t + \varepsilon_t,$$

   to estimate α (if statistically significant).
4. Treat event days as jumps:
   - Jump sizes are the ΔH_t on those dates.
   - Jump intensity $\lambda_M \approx N_{\mathrm{events}}/T_{\mathrm{years}}$.
   - Estimate $\mathbb{E}[\Delta H]$ and $\mathcal{D}_H$ (empirical distribution).
5. If α is not significantly different from zero, set $\alpha = 0$ and move to a pure jump-based accumulation model (possibly with λ_M made premium-dependent).

5.4 Calibration of BTC Price Dynamics

- Baseline: estimate constant volatility σ_S from historical BTC log returns (under $\mathbb{P}$).
- If using options, calibrate more sophisticated $\sigma_S(S_t, t)$ and jump parameters (λ_S, ν_S) under $\mathbb{Q}$.

5.5 Share-Count and Convertible Calibration

- From equity and convertible issuance disclosures, reconstruct N_t^{sh} and event-driven changes.
- Optionally model future issuance/convertible conversion behavior as a function of premium and leverage (e.g. intensities depending on π_t and $\beta_{LS}(t)$).

5.6 Indicator Computation

1. Use closed-form formulas where available:
   - IBGR (total) from $(\alpha, \lambda_M, \mathbb{E}[\Delta H])$ and H_0.
   - ILE from (H_t, S_t, D_t).
   - PMRI from OU parameters and current π_0.
   - TEE from $\gamma_{\pi S}$ and current $\beta_{LS}(t)$.
2. For IFRD, per-share IBGR, and survival probabilities, run Monte Carlo simulations under $\mathbb{P}$.

6 Implementation Plan in Python

6.1 Project Structure

A possible directory layout:

```
project_root/
  data_raw/
  data_processed/
  src/
    data_io.py
    preprocessing.py
    instruments.py
    calibration.py
    simulation.py
    indicators.py
    plots.py
  notebooks/
  results/
```

6.2 Core Modules

1. `data_io.py`

- `load_btc_prices()`: load/download BTC-USD historical prices.

- `load_mstr_prices()`: load/download MSTR prices.
- `load_btc_holdings()`: read BTC holdings and event history.
- `load_debt_schedule()`: read debt instruments.
- `load_shares_outstanding()`: reconstruct N_t^{sh}.

2. `instruments.py`

- Define a `DebtInstrument` class with maturity, coupon, face value, and convertible fields.
- Provide methods to compute outstanding face at time t and, later, effective debt.

3. `preprocessing.py`

- `build_daily_panel()`:
  - Merge BTC, MSTR, holdings, debt, and shares.
  - Compute A_t, NAV_t, $\text{NAV}_t^{\text{clip}}$, E_t, π_t, B_t.
  - Tag distress regime indicators.

4. `calibration.py`

- `fit_ou_premium(pi_series)`: estimate $(\kappa, \theta, \sigma_\pi)$ and OU diagnostics.
- `estimate_rho_and_gamma(btc_returns, premium_changes)`: estimate ρ and $\gamma_{\pi S}$.
- `fit_holding_dynamics(H_series, pi_series, event_flags)`: estimate $\alpha, \lambda_M, \mathbb{E}[\Delta H]$.
- `fit_btc_vol(btc_returns)`: estimate σ_S.

5. `simulation.py`

- Vectorized simulation of $(S_t, \pi_t, H_t, N_t^{\text{sh}}, D_t, E_t)$ under $\mathbb{P}$.
- Implementation details:
  - Construct the correlation matrix
    $$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$
    and its Cholesky factor once.
  - At each time step, draw independent standard normals and multiply by the Cholesky factor to obtain correlated (Z_S, Z_π).
  - Simulate jumps via Poisson draws and jump distributions.

6. `indicators.py`

- `compute_IBGR(params, state0)`: compute total IBGR.
- `compute_IBGR_per_share(simulated_B_paths)`: estimate $\mu_{H/N}$ from Monte Carlo.
- `compute_ILE(panel)`: compute $\beta_{LS}(t)$ over time.
- `compute_TEE(panel, gamma_piS)`: compute $\beta_{\mathrm{TOT}}(t)$.
- `compute_PMRI(params, pi0)`: compute PMRI.
- `compute_IFRD(sim_paths)`: compute IFRD distribution from pathwise $G_{\Delta t}$.
- `compute_survival(sim_paths, eps, T)`: compute survival probabilities.

7. `plots.py`

- Time series plots for $(S_t, H_t, N_t^{\mathrm{sh}}, B_t, \pi_t, \mathrm{NAV}_t, E_t)$.
- Scatter of ΔB_t vs π_t (accretion vs premium).
- IFRD histograms, leverage elasticity distributions, and survival probability curves.

6.3 Coding Checklist

1. Implement data loaders and build the daily panel.
2. Fit OU premium parameters and estimate $\rho, \gamma_{\pi S}$.
3. Estimate holding dynamics $(\alpha, \lambda_M, \mathbb{E}[\Delta H])$; decide if continuous term is necessary.
4. Estimate BTC volatility and (optionally) jump parameters.
5. Implement vectorized simulation of joint paths under $\mathbb{P}$.
6. Compute IBGR (total and per-share), ILE, TEE, PMRI.
7. Simulate IFRD and survival curves and generate plots.

7 Deliverables

- A cleaned, reproducible daily panel with $S_t, P_t, H_t, D_t, N_t^{\mathrm{sh}}, A_t, \mathrm{NAV}_t, \mathrm{NAV}_t^{\mathrm{clip}}, E_t, \pi_t, B_t$.
- A parameter report: $(\kappa, \theta, \sigma_\pi, \rho, \gamma_{\pi S}, \alpha, \lambda_M, \mathbb{E}[\Delta H], \sigma_S)$.
- A Monte Carlo engine for $(S_t, \pi_t, H_t, N_t^{\mathrm{sh}}, D_t, E_t)$ under $\mathbb{P}$.
- Indicator dashboards (notebooks/scripts) presenting:
  - Total and per-share IBGR.
  - ILE time series and current value.
  - TEE time series and current value.
  - PMRI and its history.

  - IFRD distribution and summary stats.
  - Survival probabilities vs horizon T and stress buffer ε.
  - BTC-per-share vs premium plots to visually test accretion efficiency.

- A written interpretation explaining whether current MSTR pricing implies realistic BTC accretion and leverage profiles, given historical behavior and calibrated parameters.